

The M03 laser rangefinder module features a compact size, high ranging accuracy, and suitability for various harsh environments. It can be integrated with a wide range of ranging-enabled devices, including drones, robots, and handheld laser rangefinders, among others. This module enables rapid and accurate distance measurement for the main control system; it employs a 650 nm semiconductor laser, offering a ranging resolution of 0.001 m, a measurement accuracy of ± 2 mm, and a maximum ranging range of 120 m. The module also features a UART (TTL-level) communication interface, making it a highly integrated, low-power, and lightweight ranging sensor.

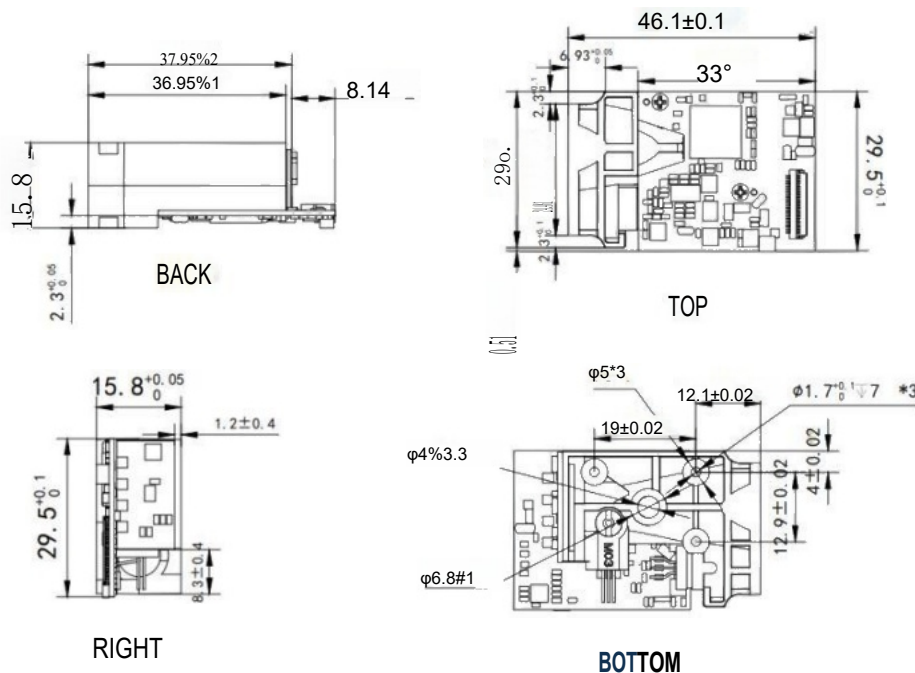
I. Module Parameters

1.1 Performance Specifications

Accuracy	$\pm (2\text{mm}+5\times 10^5\text{D})$
Unit	Millimetre
Measuring distance	0.045~120m
Measurement Time	0.9~4 seconds
Laser Classification	Class II
Laser power	<1mW
Laser wave length	620~690nm
Laser lifespan	>50000H
Operating power consumption	100mA
Standby power consumption	60mA
Size	46.1*29.5*15.8
Input voltage	3.6~4.2V
Working temperature	0~40℃
Storage temperature	-20~65℃
Weight	15G
Default baud rate	9600 bps (can be configured)

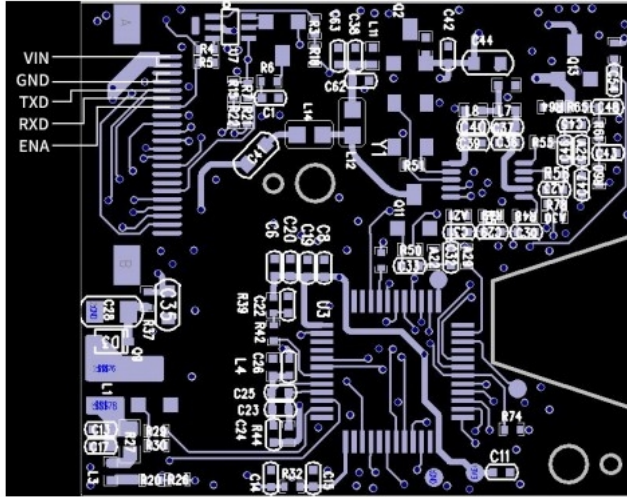
Note: Both ranging accuracy and distance are based on the assumption that the reflected target is a medium-reflectivity target; however, the actual measurement accuracy may be affected by increasing distance and variations in the reflected target, with a tolerance of $\pm(2\text{ mm} + 5\times 10^5\text{ D})$.

1.2 size



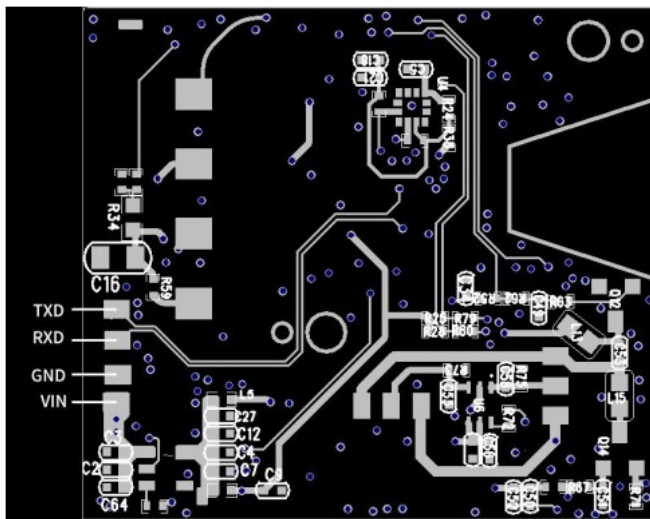
1.3 Pin

1.3.1 FPC Output Pin Definition



FPC Lead Definition (Optional)

1.3.2 Pad Pin Definition



TXD=Yellow
RXD=blue
GND=black
VIN=Red
Wiring Color Definition
(Optional)

II. Communication Protocol

- ◆ The voltage levels of the power supply and other pins must remain within the permissible range specified by the module; exceeding this range may result in permanent damage.
- ◆ When ordering this module, the external power supply parameters must be provided;
- ◆ For example, the power source for a laser rangefinder could be a lithium battery or a dry cell battery, among others.

2.1 order structure

Byte	0	1		2	3	4	5	6	7	8
Bit	[7:0]	[7]	[6:0]	[7:0]	[7:0]	[7:0]	[7:0]	[7:0]	[7:0]	[7:0]
Name	Head	R/W	Addr	Function		data	count	data		checksum
Data	0xAA	0	0x10	0x00	0x12	0x00	0x01	0x33	0x44	0x9A

The request always begins with a fixed first byte value of 0xAA; "R/W" indicates the operation: 0 – the host writes data to the slave device; 1 – the host reads data from the slave device. In this table, the slave device address is 0x10; the address is 7 bits long, with an address range of 0x00–0x7F; 0x00 is the default address for each module.

Address: 0x7F is the broadcast address;

"Function" refers to the content of the host's operational command sent to the slave device;

Data Count represents the count of data written to the slave device;

'Data' represents the data to be written to the slave device;

Checksum = R/W Address + Function Bytes + Data Count + Data.

2.2 control function

Order number	Functional Code	Name	Function
0	0x0000	FUNC_ERR_CODE	System mode
1	0x0006	FUNC_BAT_VLTG	Input voltage
2	0x000A	FUNC_HARDWAR_VER	Read hardware version
3	0x000C	FUNC_SOFTWARE_VER	Read software version
4	0x000E	FUNC_SERIAL_NUM	Read serial number
5	0x0010	FUNC_SET_ADDR	Slave Address Configuration
6	0x0012	FUNC_SET_OFFSET	Set measurement offset
8	0x0014	FUNC_SET_BAUD	Set baud rate
9	0x0015	FUNC_SET_MEAS_MAXLENGTH	Set maximum measurement range
10	0x0020	FUNC_MEAS_SINGLE	Single measurement
11	0x0021	FUNC_MEAS_CONTINUOUS	Continuous measurement
12	0x0022	FUNC_MEAS_FAST	Rapid survey
13	0x0023	FUNC_MEAS_ANGLE	Angular surveying
14	0x01be	FUNC_CTLR_LD	Laser tube on/off

2.3 Control Command Detailed Description

2.3.1 Read the latest status of the module

Bytes	0	1	2	3	4
Name	Head	RW/ADDR	Func_Code		Checksum
Data	0xAA	0x80	0x00	0x00	0x80

Read command (e.g.: aa80000080)

Slave address: 0x00

Function: Read status return information: AA 8000000001 FF FF 7F

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x80	0x00	0x00	0x00	0x01	0xYY	0xZZ	Checksum

2.3.2 Read hardware version

Bytes	0	1	2	3	4
Name	Head	RW/ADDR	Func_Code		Checksum
Data	0xAA	0x80	0x00	0x0A	0x8A

Command: (e.g., aa80000a8a; Response: AA 80000A 00012502 B2)

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x80	0x00	0x0A	0x00	0x01	0xYY	0xVV	Checksum

YYVV refers to a hardware version, such as the 2502.

2.3.3 Read software version

Bytes	0	1	2	3	4
Name	Head	RW/ADDR	Func_Code		Checksum
Data	0xAA	0x80	0x00	0x0C	0x8C

Command: (e.g., aa80000c8c; Response: AA 80000C 00012503 B5)

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x80	0x00	0x0C	0x00	0x01	0xYY	0xVV	Checksum

YYVV represents the software version, e.g.: 2503

2.3.4 Read module serial number

Bytes	0	1	2	3	4
Name	Head	RW/ADDR	Func_Code		Checksum
Data	0xAA	0x80	0x00	0x0E	0x8C

Command: (e.g., aa80000e8e; Response: AA 8000 0E 00012504 B8)

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x80	0x00	0x0E	0x00	0x01	0xSS	0xNN	Checksum

2.3.5 Read Module Input Voltage

Bytes	0	1	2	3	4
Name	Head	RW/ADDR	Func_Code		Checksum
Data	0xAA	0x80	0x00	0x06	0x86

Command: (e.g., aa 80000686)

return information

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x80	0x00	0x06	0x00	0x01	0x32	0x25	Checksum

Input voltage: 0x3225 indicates

3.2.25 V 2. 3. 6 Set module address

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x10	0x00	0x01	0x00	0xYY	Checksum

Command: (e.g., aa0000100001000112; Note: After modifying the module address, the device must be powered on again; all other corresponding commands should also be updated).

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x10	0x00	0x01	0x00	0xYY	Checksum

0xYY is a module address, with a range of 0–127; address 0x7F is the broadcast address.

2.3.7 Set module measurement offset

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x12	0x00	0x01	0xXX	0xYY	Checksum

Command: (e.g., aa0000120001006477)

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x12	0x00	0x01	0xXX	0xYY	Checksum

If ZZZY = 0x0064 (100 mm), subtract 100 mm from the measurement result; if ZZZY = 0xFF9C (–100 mm), add 100 mm to the measurement result; (Note: This setting takes effect after a system restart)

2.3.8 Laser tube activation and deactivation

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x01	0xbe	0x00	0x01	0x00	0xZZ	Checksum

Command: (e.g., aa0001be00010001c1 – Turn red light on / aa0001be00010000c0 – Turn red light off)

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x01	0xbe	0x00	0x01	0x00	0xZZ	Checksum

ZZ=0x1: Turn on the laser; ZZ=0x0: Turn off the laser tube;

2.3.9 Angle Measurement (Optional)

Bytes	0	1	2	3	4
Name	Head	RW/ADDR	Func_Code		Checksum
Data	0xAA	0x80	0x00	0x23	0xA3

return information:

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x23	0x00	0x01	0xFF	0xFF	Checksum

The positive and negative measurement results are represented using a 16-bit binary format.

If XXYY = 0064 (100), the measured value is $(100 \div 10 = 10.0^\circ)$.

If XXYY = FF9C (-100), the measured value is $(-100/10 = -10.0^\circ)$.

2.3.10 Start a Standard Survey

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x20	0x00	0x01	0x00	0x00	0x21

Command: (e.g., aa0000200001000021; returned information: AA 000020000400000484023D EB)

Bytes	0	1	2	3	4	5	6:9	10:11	12
Name	Head	RW/ADDR	Func_Code		Data count		Data	Signal	Checksum
Data	0xAA	0x80	0x00	0x20	0x00	0x04	0xAABBCCDD	0xYYZZ	Checksum

AABBCCDD are all BCD codes;

For example, if 0xAABBCCDD = 0x12345678, it represents

12345.678 m; 0xYYZZ indicates signal quality – the higher the value, the better the signal quality;

2.3.11 Start a quick measurement

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x22	0x00	0x01	0x00	0x00	0x23

Command: e.g., aa0000220001000023; Return Information: AA 000022000400000491032F ED)

Bytes	0	1	2	3	4	5	6:9	10:11	12
Name	Head	RW/ADDR	Func_Code		Data count		Data	Signal	Checksum
Data	0xAA	0x80	0x00	0x22	0x00	0x04	0xAABBCCDD	0xYYZZ	Checksum

AABBCCDD are all BCD codes;

For example, if 0xAABBCCDD = 0x12345678, it represents 12345.678

m; the value of 0xYYZZ indicates signal quality – the higher the value, the better the signal; however, the accuracy of this measurement result is slightly lower than that of a standard single measurement.

2.3.12 Enable/Disable automatic continuous measurement

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data:	0xAA	0x00	0x00	0x21	0x00	0x01	0x00	0x0Y	0x22
– Command: (e.g., aa 0000210001000022) – Upon receiving the command, the module returns Bytes 0.									
Name	Head	RW/ADDR	Func_Code		Data count		6	7	8
Data	0xAA	0x00	0x00	0x21	0x00	0x01	Data		Checksum
0x0Y = 0:	Begin continuous measurement.						0x00	0x0Y	0x22

0x0Y = 1: Stop continuous measurement.

0x0Y = 0: Enter continuous measurement mode and continuously transmit data to the host:

Bytes	0	1	2	3	4	5	6:9	10:11	12
Name	Head	RW/ADDR	Func_Code		Data count		Data	Signal	Checksum
Data	0xAA	0x80	0x00	0x21	0x00	0x04	0xAABBCCDD	0xEEFF	Checksum

Information content – single measurement;

If an error occurs during the measurement process, an error message will be returned; for details, refer to the Module Error Report.

0x0Y=1: If the device is in continuous measurement mode, the system will stop automatic continuous measurement while returning the response data. If the device is not in continuous measurement mode, only the response data will be returned.

2.3.13 Baud rate setting

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x14	0x00	0x01	0x00	0x0Y	Checksum

return information

Bytes	0	1	2	3	4	5	6	7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data		Checksum
Data	0xAA	0x00	0x00	0x14	0x00	0x01	0x00	0x0Y	Checksum

The default module baud rate is 9600;

Users can adjust the baud rate according to their needs; currently, two baud rates are available:

0x0Y = 0: Baud rate – 9600

0x0Y = 1: Baud rate – 115,200

Please restart the device to refresh it.

2.3.14 Push basic module information upon device startup.

When the module's serial port is communicating normally, the gateway will transmit the following serial port data.

Bytes	0	1	2:3	4:5	6:7	8:9	10	11
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Name	Head	RW/ADDR	Data					Checksum
Data	0xAA	0x80	0xAABB	0xCCDD	0xYYVV	0xSSNN	0x0Y	Checksum

The user can determine whether normal communication is established with the module by checking the serial port information:

0x80: RW 1 – The host reads data from the slave device; this indicates that the slave device address is 0x00;

0xAABB: Represents the PCB version number as AABB (expressed in hexadecimal).

0xCCDD: Represents the software version number CCDD (expressed in hexadecimal).

0xYYVV: Current module input voltage (represented in BCD code; 0x3225 represents 3.225 V)

0xSSNN: Module serial number.

0x0Y: Represents the baud rate configured for the current module:

0x0Y = 0: Baud rate – 9600

0x0Y = 1: Baud rate – 115,200

2.3.15 Module Error Report

Bytes	0	1	2	3	4	5	6:7	8
Name	Head	RW/ADDR	Func_Code		Data count		Data	Checksum
Data	0xEE	0x00	0x00	0x00	0x00	0x01	ERR_CODE	Checksum

2.3.16 Other error codes

ERR_CODE	Explain
0x0000 (ERR_NO)	Inerrancy
0x0001 (ERR_HARDWARE)	Hardware error
0x0003 (ERR_SIGNAL_WEAK)	The reflected signal is too weak.
0x0004 (ERR_SIGNAL_STRONG)	The reflected signal is too strong.
0x0007 (ERR_BAT_LOW)	Low power supply voltage (<3.5 V)
0x0008 (ERR_OVER_RANGE)	No to scale
0x0009 (ERR_READ)	Reading communication error
0x000A (ERR_WRITE)	Write communication error
0x000B (ERR_ADDR)	Address error

III. Factors affecting ranging capability, ranging response speed, and speed measurement accuracy

3.1 target reflectivity:

Generally, the higher the target reflectivity, the better the ranging capability and the faster the ranging response speed; for example, a target with moderate reflectivity can be measured up to 40 meters away, a target with high reflectivity can be measured up to at least 60 meters away, while a low-reflectivity target may only be measurable up to 30 meters away (for targets that are difficult to reflect light diffusely, such as water surfaces, measurement may not be possible).

3.2 Target shape:

When the reflective surface area of the measurement target is too small or when the surface is uneven, both the ranging capability and the ranging response speed will be correspondingly reduced;

3.3 Measure angle:

When the laser beam is incident perpendicularly on the reflective surface of the measurement target, the ranging capability and

ranging response speed are higher; conversely, both the ranging capability and ranging response speed decrease. Under extreme measurement angles, it cannot be guaranteed that the performance levels and ranging response speeds specified in this manual will be achieved (under special conditions, the angle between the black smooth surface shall not be less than 60°).

3.4 Measurement Environment:

Factors affecting ranging capability and ranging response speed also include the concentration of water vapor and suspended particulate matter in the air; laser reflectivity and measurement results may vary depending on environmental conditions, the color of the target, surface finish, size, actual shape, and other characteristics. Measurement errors or failures may occur under the following circumstances:

- ◆ Small or slender target
- ◆ The target is black or darker.
- ◆ The target features a graded surface.
- ◆ Target is in motion or vibration
- ◆ When measuring the water surface
- ◆ Target measured through glass
- ◆ Target: Glass or mirror surface
- ◆ The laser beam is incident at an oblique angle onto the target reflective surface.

3.5 Measure reaction time:

This product utilizes a red laser beam for measurement. It measures the time elapsed between the transmission of the laser beam from the ranging telescope to the target and its return. Maximum measurement time: 4s; target switching response time: <0.9s.

Note: Inconsistent target reflectivity and the stability of the module-locked target can affect the module's response speed.

IV. Precautions

- 4.1 When using this module, do not look directly at the laser beam.
- 4.2 Do not operate this module using the eyepiece or any other additional optical devices to avoid increasing the risk of eye injury.
- 4.3 Do not disassemble any modules; products that have been disassembled are not covered under the warranty.
- 4.4 When transporting, storing, or using this product, take care to avoid exposure to humid environments; operating the device in humid conditions—such as those prone to condensation or frost formation—may impair its ranging performance and could potentially damage the module!
- 4.5 During transportation, please add sufficient cushioning material to the packaging box to prevent damage to the module.
- 4.6 Place the module out of reach of children; do not place it on an unstable elevated surface to prevent the module from falling and sustaining damage.
- 4.7 Do not place the device inside a vehicle exposed to direct sunlight, in an environment with intense ultraviolet radiation, or near any heat source, as this may cause an uncontrollable impact on the module.
- 4.8 Under conditions of rapid temperature changes, condensation may form on the surface of the module's main lens; in such cases, do not use the module.
- 4.9 If the lens is dirty, simply wipe it clean gently with a lens cleaning cloth; do not use any other objects to wipe it, as this may scratch the lens surface.
- 4.10 This module comes with a one-year warranty or a lifetime warranty; if the issue is due to manufacturing defects, the module can be replaced free of charge; if the issue is caused by human error, the cost of repair or replacement of components will be charged based on the actual condition of the unit.

V. The range of this ranging module is defined under the following conditions:

- 5.1 The measured target reflection rate should be above 85%.
- 5.2 The measured target reflective surface is perpendicular to the laser emission direction.
- 5.3 Recommendation: When measuring distant targets, use a tripod to securely mount the module to minimize any vibration during the measurement process, thereby achieving more accurate results.
- 5.4 Any other non-standard testing environments or methodologies not described in the form shall be subject to mutual negotiation and confirmation, and both parties shall cooperate with the client to resolve any related issues. The results of such non-standard testing shall not serve as a basis for determining product quality until they have been formally confirmed.